

Improve Immersion in Virtual Reality-Based Basketball Training By Haptic Feedback

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ABSTRACT

Many studies have investigated the feasibility of using real sport equipment in VR-based sport training. However, providing haptic feedback to mimic real-world ball interaction (passing and catching) is still less explored. In this paper, a VR-based basketball training system comprising vibrotactile gloves is proposed. With the aid of haptic feedback, the immersive feelings related to interaction with a basketball in a virtual court are improved. A within-subject experiment which compared with/without haptic feedback conditions was conducted. The improvement of presence was objectively measured by an electroencephalography (EEG) approach using a portable, low-cost EEG headset in a virtual catching scenario. Furthermore, the passing reaction time was assessed in a VR-based tactic training scenario to explore the effect of haptic feedback on the performance during training. The results demonstrated the possibility of using low-cost EEG headset to assess the presence. Because the perception of catching was mainly affected by visual, haptic feedback did not make significant effect on the passing reaction time. Nevertheless, the proposed gloves was commented to improve the experience of passing and catching.

CCS CONCEPTS

• **Hardware** → **Haptic devices**; • **Computing methodologies** → **Virtual reality**.

KEYWORDS

basketball training, electroencephalography, vibrotactile gloves, virtual reality

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1 INTRODUCTION

Currently, with the aid of emerging technology, immersive VR is being widely used to increase the variety of training scenarios for athletes and other sports professionals. Compared with training in the real world, VR can develop controllable, repeatable situations that provide precise training scenarios. For example, Brault *et al.* [1] proposed a virtual simulator with a motion capture system to train and study the performance and defensive strategies of soccer goalkeepers. Covaci *et al.* [2] designed a VR system for basketball free-throw training in a CAVE-like environment. These studies focused on utilizing visual feedback to provide the user with an immersive experience. Other studies have combined VR with real sports equipment, such as cycling and batting, to improve the interaction fidelity in an attempt to mimic real-world experience [4, 6].

However, for sports that directly interact with the ball by athlete's body, it is impractical to use a real ball in immersive VR-based training. The simulation of catching, passing, and hitting with a real ball in VR might be dangerous because the user's view is limited by the display and the ball is usually dynamic. An external haptic device that can provide a natural experience in VR is needed. None of the previous studies has explored the feasibility of using a haptic device in VR-based ball sports training. Take the VR-based basketball tactical training as an example. During the training, users need to know whether they catch the virtual ball to pass the ball at correct, immediate timing. Due to the limited field of view (FOV) of the VR headset, a user cannot pay attention to every dynamic object (i.e. teammates and ball) all the time during the practice of tactics. It is thus necessary to use additional feedback other than visuals to improve the perception of the ball in the VR-based training. Small, low mass vibrotactile gloves are one of the most common ways to provide a wearable solution to increase interaction realism [7]. In this work, vibrotactile gloves composed of actuators at each finger was developed to provide cutaneous feedback which can simulate the perception of passing and catching a basketball in the virtual court.

To investigate the feasibility of introducing haptic feedback into VR-based basketball tactical training, both subjective and objective evaluations were considered in two separate sessions in this study. In session 1, an EEG-based approach was used to objectively measure the sense of presence when using vibrotactile feedback in a catch ball task. In session 2, four tactical scenarios with/without haptic feedback were used to evaluate the subjective feelings of users and the performance of passing action by questionnaires and the reaction time to pass to an open-shot teammate. The following two hypotheses were tested: (1) greater levels of immersion in haptic feedback may create more cognitive conflict, which would in turn



Figure 1: Trainee is using the proposed VR system.

be reflected in the ERP negativity amplitude; (2) the use of haptic feedback may improve the performance of the timing/correctness of passing during tactical execution.

This study makes the following contributions:

- A VR-based basketball training system with vibrotactile gloves was proposed to provide a high immersion ball interaction and improve the user's presence during the virtual training process.
- In addition to subjective questionnaires, we use a portable and low-cost EEG headset to objectively measure the user's presence in VR based on ERP. Moreover, we designed a ball passing experiment to evaluate whether haptic feedback can improve passing performance.
- Because the visual effect is reported as the main sense to perceive the catching of the ball, our study shows that the impact of the haptic feedback on the reaction time is small. However, according to the subjective questionnaires, participants agree that the vibrotactile feedback can successfully improve the experience of interacting with the virtual basketball.

2 VR BASKETBALL TRAINING SYSTEM WITH VIBROTACTILE FEEDBACK

The proposed VR basketball training system is composed of two main components, vibrotactile gloves and a virtual reality interaction system. A computer server generates a three-dimensional virtual training scene with simulated players running on a virtual court in real time according to pre-defined tactic configuration files, which is designed using a 2D basketball tactics board application. Subsequently, the generated virtual content is displayed in the VR tactical training system, which provides a high immersion tactical training scenario and an intuitive interaction menu. When a catch ball or pass ball actions arise during the tactical execution, the vibrotactile gloves give feedback necessary to simulate the interaction with the virtual basketball, which theoretically improves the degree of immersion of the user.

2.1 Virtual Basketball Training System

One of the advantages of an HMD-based VR system is that users can highly immerse and experience training at any position from different viewing perspectives. In the proposed system, users can experience tactical execution from a first person perspective, which allows them to choose to experience the tactic from the perspective of a specific offensive player. This perspective makes it possible for

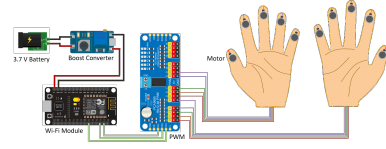


Figure 2: Proposed vibrotactile gloves.

the user to be involved in the tactical execution as would be the case on the real-world court, which provides a more immersive training experience. During the tactical execution, the user is positioned on the virtual court by a pre-defined tactical configuration file, i.e., the moving behavior and the running trajectories of the user are automatically driven by the system and the user does not need to move in the real-world. Users are free to rotate their viewing direction during tactical execution. Also, they can pass the ball to other virtual teammates in a realistic way by doing the pass ball action.

2.2 Vibrotactile Gloves

To provide haptic feedback resembling the force feedback in a catch ball and pass ball scenario in basketball, vibrotactile gloves with ten micro vibration motors and a circuit were designed, as shown in Figure 2. The device is powered by a battery, which is connected with a boost converter to generate a regulated output voltage ranging from 3.7V to 5.0V. The circuit has a Wi-Fi chip, which establishes a connection with the computing server for the purpose of receiving vibration configuration files. A vibration configuration file is a profile that defines the number of the motors to use and the vibration patterns for each motor. Based on the the pre-generated vibration configurations, pulse-width modulation (PWM) is used to control the ten micro vibration motors to give vibrotactile feedback to every finger.

In the proposed system, customized vibrotactile patterns were designed to match pass ball, hold ball, and catch ball actions. The pass and catch ball patterns were defined as a descending pulse that simulates the immediate force a player receives when interacting with the ball. The hold ball patters was defined as a small constant voltage to simulate the feeling of touching a ball.

The proposed vibrotactile gloves were composed of ten micro vibration motors(SH-MT1020), a ESP8266 Arduino Wi-Fi module, a PCA9685 PWM driver module, an MT3608 boost converter, and a 3.7V battery. The motors measured 10mm in diameter and 3mm in height.

3 EXPERIMENTS

The hypotheses are that the immersion of haptic feedback can be evaluated by mismatch negativity in ERPs and the haptic feedback may affect the timing of passing during virtual training. A within-subjects studies were conducted. The study design was approved by the local ethics committee and all participants provided informed consent before the experiment.

3.1 Apparatus

The virtual scenario was implemented using the Unity 2019.2.9 game engine and was run on an Intel Core i7-7700K CPU @ 4.20GHz,

with 16 GB RAM and Nvidia Geforce 1080 GPU. HTC Vive Pro was used as the VR display device. (two OLED displays each with a 1080 x 1200 resolution and 110° field of view). Two wrist-mounted wearable VIVE trackers were used to track the hand positions of the users.

EEG data were recorded from an Emotiv Epoc+ device with 14 sensors, which are AF3, AF4, F3, F4, F7, F8, FC5, FC6, P7, P8, T7, T8, O1, and O2. Before fitting the cap, the electrodes had to be hydrated to achieve good contact. During the data collection, the EEG data were streaming to the computing server via Bluetooth at a 256 Hz sampling rate. Experimental event markers were synchronized with the streaming data using the cortex v2 API released by Emotiv Inc.

3.2 Experimental Design

To investigate the haptic feedback in terms of the EEG signal, the subjective feelings of the users, and the timing of passing the ball, two main sessions were designed in the experiment. The first session focused on validating the potential of using ERPs to detect the prediction error in VR, which was taken as the evaluation of the presence of the haptic feedback. The second session focused on the influence of haptic feedback in terms of the participants' subjective feelings and the timing of passing the ball.

In both sessions, the participants performed each task under two various feedback conditions: (1) visual-only: the basketball was automatically attached and unattached to the virtual hands when the participants caught and passed the ball; (2) visual + vibrotactile: in addition to visual content, the participants also received vibrotactile feedback when they caught and passed the ball. The order of various conditions in both sessions was counterbalanced among the participants.

3.2.1 Session 1. EEG-based evaluation. The first session focused on the validation for the potential for using ERPs to detect the visual-haptic conflicts in a basketball VR scenario. Six participants (3 males and 3 females) were recruited from a local university campus for this session. 120 trials were presented for each feedback condition. Participants stood on a virtual court and performed a catch ball task. The trial flow is described as follows: (1) the participants moved their hands to match the position and rotation of the transparent ready hands to get ready to start the next trial; (2) After matching the ready hands for 3s, the virtual player passed the ball to the participants; (3) the participants caught the ball in 1s; (4) finally, the participants put down their hands and reset the trial.

Following the settings from the previous study [3], to understand the ERP in different visual-haptic matching conditions, participants experienced match (75%) and mismatch (25%) trials. In the matching trials, the vibrotactile feedback was presented when the ball was caught by the participants' virtual hands, which were fully in contact with the ball as in a realistic situation in which a ball is caught. In the mismatching trials, the vibrotactile feedback was presented before the ball reached the participants' virtual hands. The virtual basketball was attached with the hands in an offset position, which is an unrealistic catch ball situation. The visual-mismatch condition was implemented by enlarging the trigger capsules of the virtual hands by 200%.

3.2.2 Session 2. Subjective questionnaire and reaction time. The second session focused on an investigation of the impact of a haptic device on offensive possession of the basketball (including catching, holding, and passing) by evaluating the subjective questionnaire and the timing of passing the ball. Thirty participants (17 males and 13 females) were recruited from a local university campus for this session. These participants were graduate and undergraduate students ranging in age from 18 to 24 ($M=21.10$, $SD=1.63$) who were on their department basketball teams. Ten trials were presented for four different tactics. Participants experienced the scenario from a passer perspective, where they pass the ball to a teammate with an open shot at the last step of the tactical execution. There is a correct timing for passing to obtain a good offensive opportunity. That is, if the passer pass the ball too early, his/her teammate may not be prepared to catch the ball; if the passer pass the ball too late, the open shot opportunity may be destroyed by the defender. Following [3], the four most relevant questions (G1, REAL2, SP4, INV1) from the standard IPQ [8] were presented after each condition. Furthermore, an interview was held to understand the subjective feelings of the participants. The interview includes two questions: (1) Which is the main feedback (visual, vibration, or others) that makes you feel you catch the ball? (2) Did the vibration feedback improve the perception of holding a ball?

3.3 EEG Analysis

The data was processed by EEGLAB¹ and ERPLAB² toolboxes in MATLAB for further analysis. Raw signals were 0.16-Hz high-pass filtered and a 50-Hz low-pass filtered by a finite impulse response (FIR) filter. Subsequently, signals were epoched from 300 ms from the onset of the catching event to 700 ms after the event. To reject the data with artifacts (e.g. eye blink and muscle activity), visual inspection and independent component analysis (ICA)[5] were applied. Finally, the mean difference ERP wave was calculated to investigate the difference between match and visual-haptic conflict conditions as shown in Figure 3. Furthermore, following [3], the mismatch negativity was represented by the large negative deflection in the specific interval. We extracted the ERP negativity peaks and latencies between 100 ms to 300 ms after using a 10-Hz low pass filter to further analyze the visual-haptic conflict.

4 EXPERIMENTAL RESULTS

4.1 Session 1. EEG-based evaluation

Emotiv Epoc+ is a low-cost, portable EEG device with only 14 electrodes, which does not contain the common electrode FCz investigated in previous studies. Therefore, we observed the ERP waves at the nearby frontal and central electrodes and found an obvious negative deflection in the electrodes AF3 and AF4. As shown in Figure 3, there are strong amplitude growing as a virtual ball was caught by the user in the virtual court. The minimum negativity peaks and latencies at AF3 and AF4 were shown at 1. Both electrodes AF3 and AF4 have a greater negative amplitude for visual+vibrotactile conditions.

¹<https://sccn.ucsd.edu/eeqlab/index.php>

²<https://erpinfo.org/erplab>

	AF3		AF4	
	AMP	LAT	AMP	LAT
Visual	-0.942	222.660	-1.070	230.470
Vibra+Visual	-1.976	296.880	-2.395	261.720

Table 1: Minimum negativity peaks (μV) and latencies (ms) in 100-300ms interval of the match-mismatch difference wave at electrode AF3 and AF4 for both conditions.

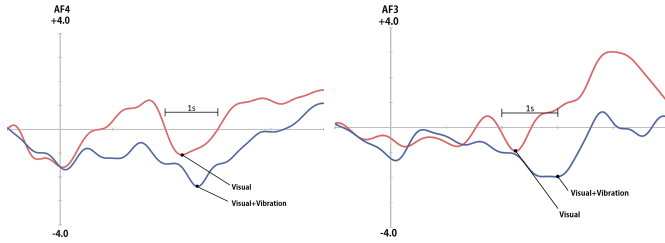


Figure 3: Amplitude of the ERPs obtained by subtracting the mean amplitude of all match trials from the mean amplitude of all mismatch trials for both feedback conditions (Visual and Vibration) at AF3 and AF4.

4.2 Session 2. Subjective questionnaire and reaction time

The histogram and table of the reaction times for each tactic were shown in Table 2. We used pair-measures t-test to statistically analyze the reaction time. Results showed that there was no significant difference between two feedback conditions. The histogram and table of four questions in the IPQ were shown in Figure 4. For the subjective self-reported presence, we used pair-measures t-test to statistically analyze the data. The results showed that there was a significant statistically difference for the REAL2, $t(30) = -2.419, p = .016$. Most of the participants (87%) voiced that the visual perception is the main factor for feeling the catching of a ball. However, over half of participants agreed that the vibration feedback highly improves the perception of catching the ball.

4.3 Discussions

Mismatch negativity could be observed at some electrodes, which showed the possibility to use EEG-based immersion evaluation in a more complicated virtual reality scenario with haptic feedback. Besides, rather than using a bulky EEG system, we used a low-cost and portable EEG device to record the raw signals during the VR experience. The results support the use of a portable EEG device to apply the visual-haptic detection technique in the automatic content adjusting based on the user's ERPs.

For the effectiveness of using haptic feedback in the tactic training, there is no significant difference between the mean reaction time for all tactics. Since most of the participants voiced that they sensed the catching of a ball by visual, the timing/correctness of passing may mainly be affected by visual perception. That is, the haptic feedback only acts as complementary feedback which does not make a great impact on the feeling of processing the ball and the learning of passing timing.

The question of REAL2 is "How much did your experience in the virtual environment seem consistent with your real world experience?", which is the only question showed a statistically higher difference between conditions. These results are consistent with the result of the interview. Even visual perception is the most important sensing for a person in the virtual environment, the introduction of haptic feedback can provide a more realistic physical interaction and improve the immersion in VR-based ball sports.

There are some limitations of this study. One is that the EEG signal was processed and analyzed in offline condition. Further evaluation in real-time applications using portable EEG device of ERPs presence was needed. The other is that the improvement of using haptic feedback in VR-based basketball training effectiveness is small. Because user mainly relied on visual perception to sense the catching of virtual basketball, haptic feedback did not improve the passing reaction time. In the future, we will evaluate dribbling in VR-based basketball training. For a better dribbler, players should not look at the ball during dribbling. For dribbling a ball without visually sensing the ball in VR, haptic feedback might be a better solution to improve the experience.

	Tactic1	Tactic2	Tactic3	Tactic4
Visual	4.11	2.70	3.72	2.83
Vibra+Visual	4.18	2.75	3.73	2.81

Table 2: Mean of the reaction time for each tactic for each condition.

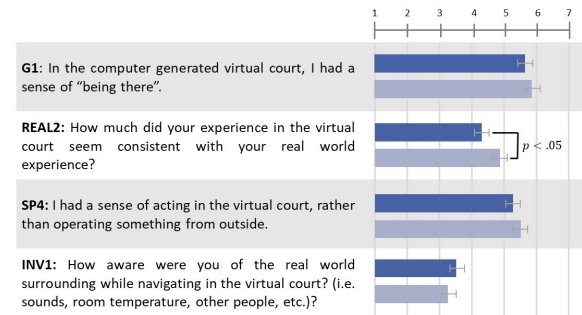


Figure 4: Histogram of the means and standard deviations for the shortened IPQ for each condition.

5 CONCLUSIONS

In this paper, we introduced the vibrotactile feedback into the VR-based basketball training to improve the experience when user catching/holding a basketball in the virtual scene. Using the EEG-based evaluation, the results showed the possibility to detect visual-haptic mismatch using low-cost device. In addition, the subjective evaluation results showed that this vibrotactile feedback can improve the experience of interacting with the virtual basketball. In the future, more participants need to be recruited in the future to validate the suitability and the electrode to observe using low-cost, portable EEG device to evaluate a haptic feedback in the VR scenario.

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